ECON 5345 Homework 8 Report

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(a) Combining the first three constraints, we have

$$Z_t K_{t-1}^{\alpha} L_t^{1-\alpha} = C_t + [K_t - (1-\delta)K_{t-1}] + \frac{\phi}{2}\left(\frac{K_t}{K_{t-1}} - 1\right)^2 K_{t-1}. \tag{1}$$

Then the maximization problem is

$$\max_{C_t, K_t, L_t} \mathbb{E} \sum_{t=0}^{+\infty} \beta^t \left\{ \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\xi L_t^{1+1/\eta}}{1+1/\eta} \right\}, \tag{2}$$

subject to constraint (1).

The Lagrangian is

$$\begin{aligned} \mathcal{L} = \mathbb{E} \sum_{t=0}^{+\infty} \beta^t \Bigg\{ & \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\xi L_t^{1+1/\eta}}{1+1/\eta} \right] \\ & + \lambda_t \left[Z_t K_{t-1}^{\alpha} L_t^{1-\alpha} - C_t - [K_t - (1-\delta)K_{t-1}] - \frac{\phi}{2}\left(\frac{K_t}{K_{t-1}} - 1\right)^2 K_{t-1} \right] \Bigg\}, \end{aligned} \tag{3}$$

where λ_t is the attached Lagrange multiplier. Taking FOCs with respect to C_t, K_t, and L_t, we get

$$[C_t]: \quad C_t^{-\sigma} = \lambda_t, \tag{4}$$

$$\begin{aligned} [K_t]: \quad & \lambda_t \left[\phi \left(\frac{K_t}{K_{t-1}} - 1 \right) - 1 \right] \\ & + \beta \mathbb{E}_t \lambda_{t+1} \left\{ \alpha \frac{Y_{t+1}}{K_t} + (1-\delta) + \frac{\phi}{2} \left[1 - \left(\frac{K_{t+1}}{K_t} \right)^2 \right] \right\} = 0, \end{aligned} \tag{5}$$

$$[L_t]: \quad -\xi L_t^{1/\eta} + \lambda_t (1-\alpha) \frac{Y_t}{L_t} = 0. \tag{6}$$

(b) Without stochasticity, we normalize $Z = 1$. Denote X^* as the steady state value of a variable X. We need to solve for $(C^*, K^*, L^*, I^*, Y^*)$, and the prices (R^*, W^*)

where R^* is the capital rental rate and W^* is the wage rate. Substituting (4) into (5) and (6), together with the constraints and the expressions for capital rental rate and wage rate, we get the system:

$$\begin{aligned} R^* + (1-\delta) &= \frac{1}{\beta}, & (7)\\ \xi(L^*)^{1/\eta}(C^*)^{\sigma} &= W^*, & (8)\\ Y^* &= (K^*)^{\alpha}(L^*)^{1-\alpha}, & (9)\\ Y^* &= C^* + I^*, & (10)\\ I^* &= \delta K^*, & (11)\\ R^* &= \alpha\frac{Y^*}{K^*}, & (12)\\ W^* &= (1-\alpha)\frac{Y^*}{L^*}, & (13) \end{aligned}$$

where (10) is derived by plugging the LoM. With 7 variables and 7 equations, the systme is just-solvable.

(c) Denote $\hat{x}_t = \frac{X_t - X^*}{X^*}$. Combining (4) and (5), we can log linearize the derived FOC of K_t to get

$$\mathbb{E}_t\hat{c}_{t+1} + \beta\left[\hat{k}_t[(\alpha-1)R^* + (1-\delta)K^* + \phi] - \mathbb{E}_t\hat{k}_{t+1} + (1-\alpha)R^*\mathbb{E}_t\hat{\ell}_{t+1}\right] = \phi(\hat{k}_{t-1} - \hat{k}_t), \tag{14}$$

where (7) is used to substitute $R^* + (1-\delta)$.

Combining (4) and (6), we can log linearize the derived FOC of L_t to get

$$(1 + 1/\eta)\hat{\ell}_t = -\sigma\hat{c}_t + \hat{y}_t. \tag{15}$$

The production function can be log linearized as

$$\hat{y}_t = \hat{z}_t + \alpha\hat{k}_t + (1-\alpha)\hat{\ell}_t. \tag{16}$$

The consumption-investment equation can be log linearized as

$$\hat{y}_t = \hat{c}_t\frac{C^*}{Y^*} + \hat{\imath}_t\frac{I^*}{Y^*}. \tag{17}$$

The LoM can be log linearized as

$$\hat{k}_t = (1-\delta)\hat{k}_{t-1} + \hat{\imath}_t. \tag{18}$$

The prices can be log linearized as

$$\hat{r}_t = \hat{y}_t - \hat{k}_t, \tag{19}$$

$$\hat{w}_t = \hat{y}_t - \hat{\ell}_t. \tag{20}$$

The productivity can be log linearized as

$$\hat{z}_t = \rho \hat{z}_{t-1} + \varepsilon_t \tag{21}$$

Equation (16) can be used to estimate α. Equation (18) can be used to estimate δ. Equation (14) can be used to estimate ϕ given the estimated α and δ.

Equation (21) can be used to estimate ρ and σ_ε.

Equation (15) can be used to estimate η and σ given the estimated σ.

(d) Yes. They match well with the data. For ϕ, combining (14) and (16), we can use data on consumption, output and labour series with one-step expectation to estimate ϕ.